Why I care about SMC

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I Don't

Bayesian inference

Suppose we have some data $\mathbf{X} = X_1, X_2 \ldots X_n$.

We also have a model that relates our data to some model parameters via a likelihood function $f(x|\theta)$. We have some prior probability distribution $\pi_0(\theta)$ encoding our a priori belief about where/how θ might be distributed.

We can encapsulate our belief about the parameter in the posterior probability distribution:

$$\pi(\theta|\mathbf{X}) = \frac{\pi_0(\theta) \cdot f(\mathbf{X}|\theta)}{\int \pi_0(\theta) \cdot f(\mathbf{X}|\theta) d\theta}$$

Things we might be interested in

The probability density

This posterior probability density we saw is in general intractable, because we don't (currently) have any idea about the integral term in the denominator.

The marginal likelihood

The denominator term is a really useful thing to have because it quantifies the *marginal likelihood* or *model evidence*. This essentially tells us how well our model fits the data.

$$p(\mathbf{X}|\mathcal{M}) = \int_{\theta \in \Theta} f(\mathbf{X}|\theta, \mathcal{M}) \cdot \pi_0(\theta|\mathcal{M}) d\theta$$

The posterior mean

$$\mathbb{E}_{\theta \sim \pi(\cdot|\mathbf{X})}[\theta]$$

Other things we might be interested in

The posterior variance

$$\mathbb{E}_{\theta\sim\pi(\cdot|\mathbf{X})}[\theta^2] - \left(\mathbb{E}_{\theta\sim\pi(\theta|\mathbf{X})}[\theta]\right)^2$$

Extreme value probabilities

$$\mathbb{P}(\theta > c|\mathbf{X}) = \mathbb{E}_{\theta\sim\pi(\cdot|\mathbf{X})}[\mathbb{I}(\theta > c)]$$

Expectations

In general we want to evaluate expectations wrt the posterior .

Another thing we might want

Posterior predictive distribution

$$\mathbb{P}(y|\mathbf{X}, \mathcal{M}) = \int_{\Theta} f(y|\theta, \mathcal{M}) \cdot \pi(\theta|\mathbf{X}, \mathcal{M}) d\theta$$

Monte Carlo Integration

Suppose we have N i.i.d. samples $\theta_1, \ldots \theta_N$ from a distribution with density $f(\theta)$.

By the law of large numbers, we can get a consistent estimator of the expectation of any function $a(\theta)$ w.r.t. f.

By the law of large numbers, as $n \to \infty$:

$$\frac{1}{N} \sum_{i=1}^{N} a(\theta_i) \to \mathbb{E}_{\theta \sim f(\cdot)}[a(\theta)]$$